IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE PART II: MEng, BEng and ACGI

Corrected Copy

DEVICES

Tuesday, 28 May 10:00 am

Time allowed: 1:30 hours

There are TWO questions on this paper.

Answer ALL questions. Question One carries 25 marks and Question Two carries 25 marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : W.T. Pike
Second Marker(s) : Z. Durrani

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Special instructions for invigilators

Special instructions for students

Do not use red nor green ink.

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Constants

permittivity of free space:	$\varepsilon_o = 8.85 \times 10^{-12}$ F/m
permeability of free space:	$\mu_o = 4\pi \times 10^{-7}$ H/m
intrinsic carrier concentration in Si:	$n_i = 1.45 \times 10^{10}$ cm^{-3} at $T = 300$ K
dielectric constant of Si:	$\varepsilon_{Si} = 11$
dielectric constant of SiO_2:	$\varepsilon_{ox} = 4$
thermal voltage:	$V_T = kT/e = 0.026$ V at $T = 300$ K
charge of an electron:	$e = 1.6 \times 10^{-19}$ C
Planck's constant:	$h = 6.63 \times 10^{-34}$ Js
bandgap of Si:	$E_G = 1.12$ eV at $T = 300$ K
electron affinity of Si	$\chi = 4.05$ eV at $T = 300$ K
effective density of states of Si:	$N_C = 3.2 \times 10^{19}$ cm^{-3} at $T = 300$ K $N_V = 1.8 \times 10^{19}$ cm^{-3} at $T = 300$ K

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Formulae

$$\left.\begin{aligned} J_n(x) &= e\mu_n n(x)E(x) + eD_n \frac{dn(x)}{dx} \\ J_p(x) &= e\mu_p p(x)E(x) - eD_p \frac{dp(x)}{dx} \end{aligned}\right\}$$ Drift-diffusion current equations

$$\left.\begin{aligned} \frac{\partial \delta n}{\partial t} &= D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n} \\ \frac{\partial \delta p}{\partial t} &= D_p \frac{\partial^2 \delta p}{\partial x^2} - \frac{\delta p}{\tau_p} \end{aligned}\right\}$$ Continuity equations of minority carriers

$$\left.\begin{aligned} J_n &= \frac{eD_n n_p}{L_n}\left(e^{\frac{eV}{kT}} - 1\right) \\ J_p &= \frac{eD_p p_n}{L_p}\left(e^{\frac{eV}{kT}} - 1\right) \end{aligned}\right\}$$ Text-book diode diffusion currents

$$V_0 = \frac{kT}{e}\ln\left(\frac{N_A N_D}{n_i^2}\right)$$ Built-in voltage

$$c = c_0 \exp\left(\frac{eV}{kT}\right) \text{ with } \begin{cases} c = p_n \text{ or } n_p \\ c_0 \text{ bulk minority carrier concentration} \end{cases}$$ Minority carrier injection under bias V

$$\delta c = \Delta c \exp\left(\frac{-x}{L}\right) \text{ with } \begin{cases} \delta c = \delta p_n \text{ or } \delta n_p \\ \Delta c \text{ the excess carrier concentration} \\ \text{at the edge of the depletion region} \end{cases}$$ Excess carrier concentration as a function of distance when recombination occurs – long layer approximation.

$$L = \sqrt{D\tau}$$ Diffusion length

$$D = \frac{kT}{e}\mu$$ Einstein relation

$$C_{diff} = \frac{e}{kT} I\tau$$ Diffusion capacitance

$$i(t) = \frac{Q(t)}{\tau} + \frac{dQ(t)}{dt}$$ Time variation of current and charge

$$\frac{dE}{dx} = \frac{\rho(x)}{\varepsilon}$$ Poisson equation in 1 dimension

$$n = N_C \exp\left(\frac{E_F - E_C}{kT}\right)$$

$$p = N_V \exp\left(\frac{E_V - E_F}{kT}\right)$$ Carrier concentrations

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1. High voltage pn diodes.

a) Explain how and when avalanche breakdown occurs in a diode, and its effects. [5]

b) The maximum electric field, E_{max}, within a diode as a function of the voltage, V, across the diode can be derived from Poisson's equation as:

$$E_{max} = \sqrt{\frac{2eN_AN_D(V_0 - V)}{\varepsilon(N_A+N_D)}}$$

where V_0 is the built-in voltage. Where in the device is this field a maximum? Show that for a p^+n diode the reverse voltage is limited by the avalanche breakdown field, E_{BD}, to a maximum of

$$V \approx -\frac{\varepsilon E_{BD}^2}{2eN_A}$$

and calculate the doping level to allow at least a 100 V reverse voltage for a breakdown field of 3×10^5 V/cm. Is the actual voltage likely to be larger or smaller, and why? [10]

c) Explain why for a p^+n diode used in a switching power application a lightly doped, n- layer is introduced, and give two reasons why it is n- rather than intrinsic. [5]

d) Explain "punch through" and how the width of the n^- layer sets the reverse voltage at which it occurs? [5]

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2. A cross section of a generic, vertical-geometry power device is shown in fig. 2.1, with the critical device regions and contacts outlined but unlabelled.

a) Draw two copies of this cross section and label all the contacts and the doping of each device region for i) an n-channel DMOS (diffused metal-on-insulator) power device, ii) a corresponding IGBT (insulated gate bipolar transistor) power device. [5]

b) For each of these two devices, explain how the current through the device is controlled. [5]

c) For the DMOS, what predominantly determines the ON resistance, and how does this change in the IGBT? [5]

d) For each of these two devices, what limits the switching speed, and which device is faster? [5]

e) In the light of above considerations, which of the two devices would be more appropriate for use in the battery power management of an electric vehicle? Also identify one further device parameter that will influence the choice. [5]

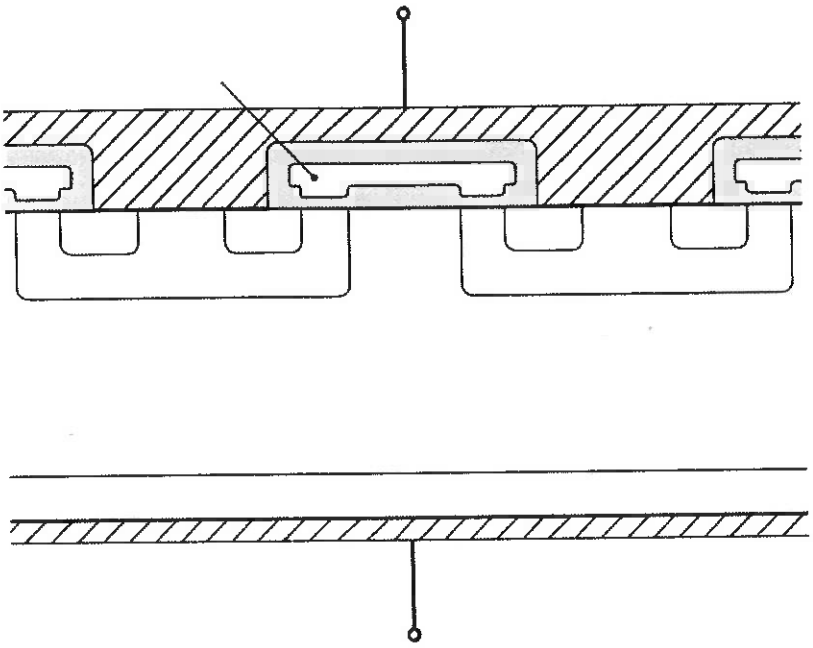


Figure 2.1: Cross sections of a generic vertical-geometry power device.

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